

The background is white and decorated with various colorful geometric shapes and patterns. There are pink and blue circles, a yellow rectangle with black dashed lines, a green zigzag line, a blue grid of dots, and several small black dots arranged in patterns. The overall style is modern and playful.

VCE Methods

Tips and Advice

Jesse M, 2025

Introduction

My name is Jesse Mogg, and I graduated in 2024 achieving a raw methods study score of 49.

Study	Year Completed	Unit 3 Result	Unit 4 Result	Grade 1	Grade 2	Grade 3	Study Score
MA11	2024	S	S	A+	A+	A+	49

My exam scores:

Exam 1 – 37/40 marks: **Final Score 74 out of a maximum of 80.**

Exam 2 – 79/80 marks: **Final Score 118 out of a maximum of 120 for this component.**

I am currently doing a Bachelor of Science at Melbourne Uni – likely majoring in computer science.

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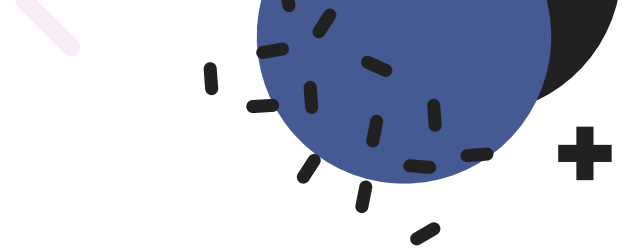
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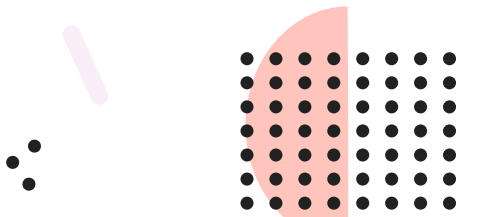
Question Approach



01

Study Techniques

What worked for me



The Basics

- Take notes.
- Do the assigned practice questions.
- Reach out to your teachers for explanations.
- Take time to ensure you have at least a basic understanding of the vast majority of the topics.
- Try to establish a consistent study schedule – **build good habits** .

Practice Exams

The questions you get in the textbooks are really nothing like what you'll see in actual assessments.

What you actually see, is that exam questions almost always follow the same format of every other exam – they're kind of predictable.

The more exams questions you're exposed to, the more you'll get used the kinds of concepts and tricks examiners are trying to throw at you.

Therefore, you want to do as many practice exams, SACs, questions, etc as you possibly can leading up to an assessment.

- Don't start doing this the week before the exam, **start as early as possible** .
- If you're not confident right now, do untimed – meticulous and precise. If you are confident, do exam-condition.

Reflecting on Mistakes

For trivial mistakes (arithmetic, etc), you don't have to do this, but if you find a question you genuinely can't do:

- take the time to figure out how it's meant to be done – don't just skip it.
- take note of both the solving technique, and what question it is.

You'll basically build up a big list of intricate techniques to solve difficult questions – of which the majority of students across the state probably can't do.

Every so often, **revise this list** . Visit those questions again if you can.

- If you're strongly confident that you know exactly how it's meant to be done, great!
- If not, **do it again** . Practice will force you to remember.

Reflecting on Mistakes

All of my scanned notes: <https://drive.google.com/drive/folders/1v47xbP1fpQwworDA8DEqroDRTDN7IAEI>

Gradients with restricted domains 2019 Exam 2 SA Q26

The gradient at a point is only defined if it can be approached from the left and right. This means the derivative at function endpoints is not defined.

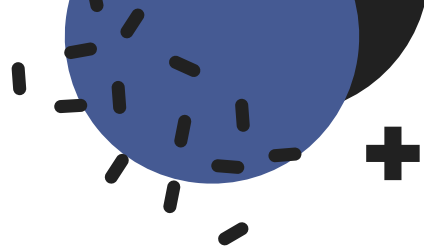
Involutorial VDF working 2019 Exam 2 SA Q48ii

For questions involving the use of the 'involutorial' VDF with ≥ 2 marks, we must give the value that leads up to the desired result.

This is displayed in the VDF's output.

E.g. $Pr(X \geq 6) = 0.0107$, $Pr(X \geq 7) = 0.0024$ (1% is threshold).

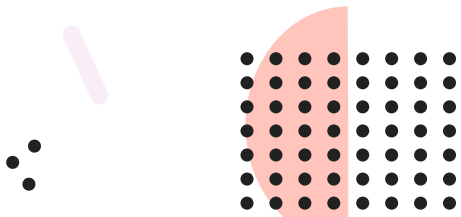
$\therefore n = 7$ is the solution



02

User Defined Functions

(UDFs)



What are UDFs?

UDFs are custom CAS functions that are specifically created to streamline typical methods concepts.

This means they can **significantly speed up** trivial questions and present all information needed in what is essentially a single calculation.

They live in the CAS custom library (6th tab in the book) and can be accessed within any working document.



▼ Average Rate of Change (avgroc)

Function: Determines the average rate of change of a function

Syntax: avgroc(function, variable, lower, upper)

Example:

$$\text{Let } f: [-1, \infty) \rightarrow \mathbb{R}, f(x) = x^2 - 3\sqrt{x+1}$$

Determine the average rate of change of $f(x)$ from $x = 0$ to $x = 5$

$$\text{avgroc}(x^2 - 3\sqrt{x+1}, x, 0, 5)$$

► Average Rate of Change:

► Working:

Interval: $0 \leq x \leq 5$

$$\text{Let } f(x) = x^2 - 3\sqrt{x+1}$$

$$\frac{f(5) - f(0)}{5 - 0}$$

$$\frac{25 - 3\sqrt{6} - (-3)}{5}$$

► Answer:

$$\frac{-3\sqrt{6} - 28}{5}$$

Question 4

Consider the system of simultaneous linear equations below containing the parameter k .

$$\begin{aligned} kx + 5y &= k + 5 \\ 4x + (k+1)y &= 0 \end{aligned}$$

The value(s) of k for which the system of equations has infinite solutions are

- A. $k \in \{-5, 4\}$
- B. $k \in \{-5\}$**
- C. $k \in \{4\}$
- D. $k \in \mathbb{R} \setminus \{-5, 4\}$
- E. $k \in \mathbb{R} \setminus \{-5\}$

from mm_func

$$\text{linesolve}(k \cdot x + 5 \cdot y = k + 5, 4 \cdot x + (k + 1) \cdot y = 0)$$

► Answer:

$$\begin{bmatrix} \text{"Unique"} & k \neq -5 \text{ and } k \neq 4 \\ \text{"None"} & k = 4 \\ \text{"Infinite"} & k = -5 \end{bmatrix}$$

Approximation Algorithm UDFs

Newton's Method -
mm_calc



```
newton(Expr,Var,x0,n)
```

Trapezium Area
Approximation - mm_calc



```
trapapprox(Expr,Var,Low,Up,n_trap)
```

Bisection Method -
mm_func



```
bisec(Expr,Var,x0,x1,lter)
```

Question 13

The following algorithm applies Newton's method using a **For** loop with 3 iterations.

```
Inputs: f(x), a function of x
          df(x), the derivative of f(x)
          x0, an initial estimate

Define newton(f(x), df(x), x0)
  For i from 1 to 3
    If df(x0) = 0 Then
      Return "Error: Division by zero"
    Else
       $x0 \leftarrow x0 - f(x0) \div df(x0)$ 
  EndFor
  Return x0
```

The **Return** value of the function $\text{newton}(x^3 + 3x - 3, 3x^2 + 3, 1)$ is closest to

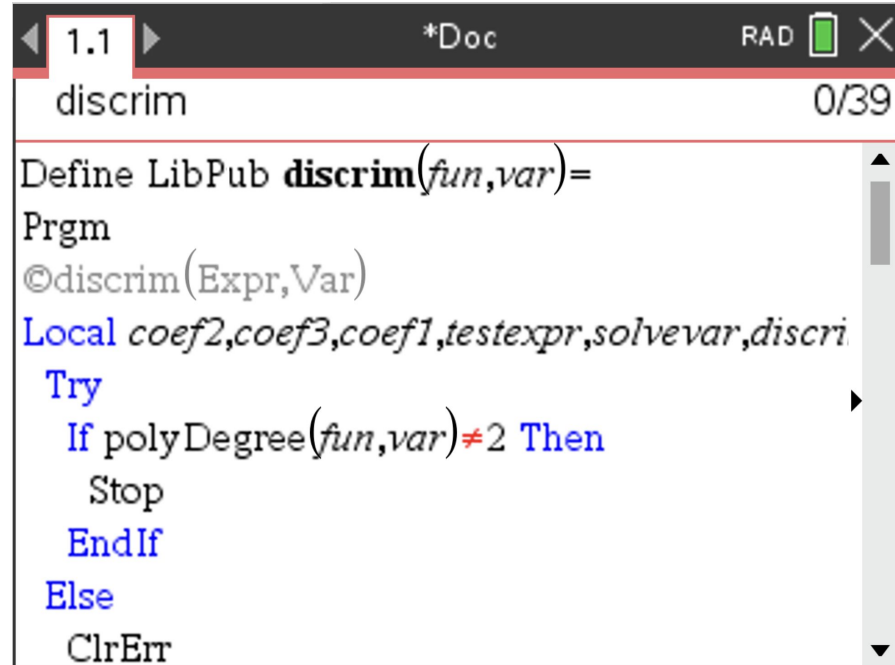
- A. 0.83333
- B. 0.81785
- C. 0.81773
- D. 1
- E. 3

*mm_calc*newton($x^3 + 3 \cdot x - 3, x, 1, 3$)

"n"	"xn"	"f(xn)"
0.	1.	1.
1.	0.833333	0.078704
2.	0.817851	0.000596
3.	0.817732	3.47177E-8

Inspecting UDF Code

We can actually look at the code used in the UDFs to see how they work.



```
1.1 *Doc RAD 0/39  
discrim  
Define LibPub discrim(fun,var)=  
Prgm  
@discrim(Expr,Var)  
Local coef2,coef3,coef1,testexpr,solvevar,discrim  
Try  
If polyDegree(fun,var)≠2 Then  
Stop  
EndIf  
Else  
ClrErr
```

mm_func\discrim

UDFs aren't perfect

Typically slower than by hand or regular CAS

intsolve
ccondpr
dcondpr
prtable
samplebinompr
pcheck

UDFs are powerful, but they aren't perfect. There are clearly some that are more practical than others.

Though, even the best UDFs aren't always the most appropriate for a given question.

In the same way some exam 2 multichoice are better solved without a CAS, often UDF applications can be unnecessary or less efficient.

UDFs sometimes break, produce misleading results, or in some cases crash the CAS.

It should not be the only way you know to solve a question – you should understand how it does it what it does.

I never found use for them

nroot
nstp
npoi
samplehypergeo
samplehyppr
signtab

Awkward to use

transform



Jesse MOGG

Sun, Nov 3, 2024, 3:28 PM



Hey LazyMath, I think I've found a bug with `mm_calc\boundarea`. For the 2023 SH exam, for question 5f, using the UDF yields the correct endpoints and expression,



LazyMath <lazymath2024@gmail.com>

Sun, Nov 3, 2024, 4:00 PM



to me ▾

Hey Jesse,

Thanks for raising this. I've fixed it. It was just a small error of me leaving a function out of the case. Accidentally did $-g(x)$ rather than $f(x) - g(x)$:(

What to do now

1. Visit <https://lazymath.github.io/> and look at the FAQs to find out how to install them.
2. Print out the guide PDF and stick it in the back of your bound reference.
3. Progressively familiarise yourself with them. You'll come across certain questions and instinctively recognise a particular UDF will help you solve it quickly.

Also feel free to let LazyMath know about any bugs you find, they usually patch them quite quickly.

I also highly recommend signing up to their mailing list

Subscribe for Updates →



03

Maximising Efficiency

General Exam Advice

Some things that I found work quite well for me are:

- Going through short-answer first, then multichoice (exam 2, CAS)
- Writing an asterisk (*) next to questions I couldn't quickly begin to answer.
- Writing ticks (✓) next to questions I feel I've properly answered.
- Taking time with working out to avoid silly arithmetic mistakes.
- Clearly stating the justifications that I expect examiners to be looking for in their markings – more on this later.
- Re-reading the question to ensure I've answered **all aspects** of the question.

Deducing Mark Allocation

The more questions you do and mark yourself on, the more you'll begin to understand exactly what is expected from a given type of question. Again, it's just practice.

Try to think about what the examiner are actively looking for in your solution – they are most likely looking for one-line statements that prove you know what you're doing. For example, what I can deduce will be the mark allocation:

- a. State the coordinates of all axial intercepts of f .

1 mark

the coordinates of the axial intercepts listed – one mark.

- b. Find the coordinates of the stationary points of f .

2 marks

how you're able to get the stationary points of f – one mark. ($f'(x) = 0$)

the coordinates of the stationary points listed – one mark.

Building **CAS** Skill

Getting good with the CAS is **extremely** important – around ~65% of your final year 12 assessment score is with CAS mandated.

Ideally you want to get good at:

- Inputting expressions with the CAS – i.e. using special keybinds to do things quickly.
- Graphing functions and using sliders to observe behaviour as variables change.
- Knowing many CAS functions – native and UDFs.
- Recognising when certain calculations have potential to crash the CAS.

Problem Pages

You should create problem pages every time you start working on a new question with new functions, variables, etc.

E.g. these two questions:

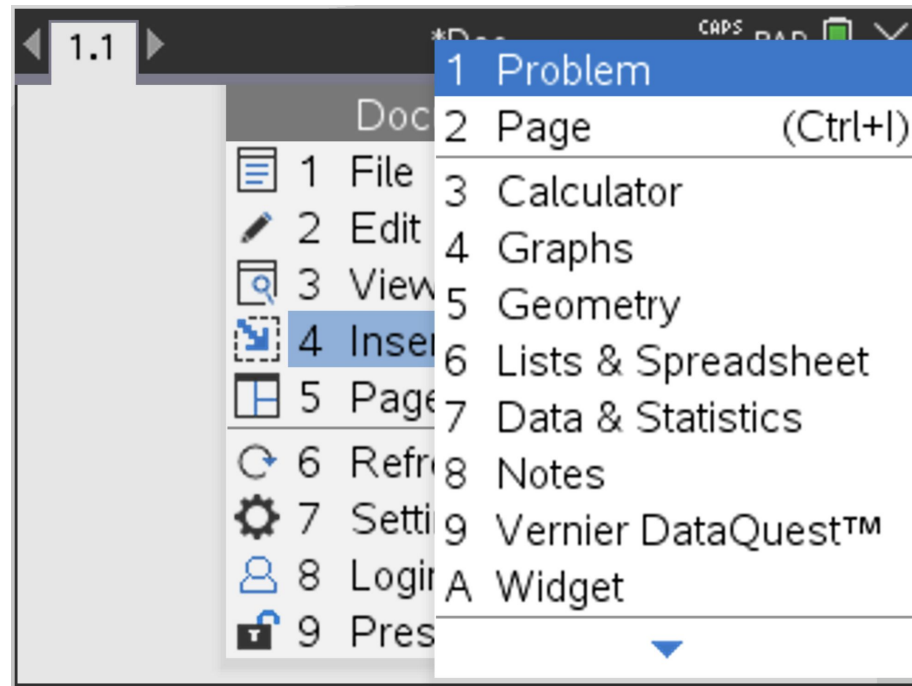
Question 1 (10 marks)

Consider the function $f: R \rightarrow R, f(x) = x^3 - px$, where $p \in R$.

Question 5 (12 marks)

Let $f: R \rightarrow R, f(x) = e^{-ex}$.

should have their own problem pages, each with their distinctly defined $f(x)$.



doc ▾ ————— Insert ————— Problem

Non-CAS Exam 2 Questions

You'll find questions on exam 2 that the CAS is either unable to do, or is inefficient.

Sometimes you should think through a problem from a non-CAS perspective, and you might recognise that you don't need to do any actual calculations.

Question 16

The values for two continuous functions, f and g , and their derivatives are given in the tables below.

	$x = 0$	$x = 2$
$f(x)$	2	-1
$f'(x)$	-1	2

	$x = 0$	$x = 2$
$g(x)$	1	-1
$g'(x)$	0	3

What is the value of $\frac{d}{dx}((g \circ f)(x))$ at $x = 0$?

A. -3

B. -1

C. 0

D. 1

E. 2

Apply chain rule by hand:

$$f'(x) * g'(f(x))$$

At $x = 0$:

$$f'(0) * g'(f(0)) = (-1) * g'(2) = (-1) * (3) = -3$$

Non-CAS Exam 2 Questions

1. Let $f(x) = 9 - x^2$. Find the area bounded by $y = f(x)$ and the positive axes.

$$\int_0^3 f(x) \, dx$$

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2. Find the expression for $f(3x)$.

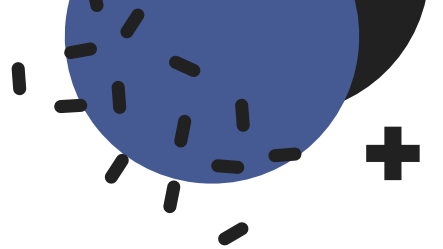
$$f(3 \cdot x)$$

$$9 - 9 \cdot x^2$$

3. Find the area bounded by $y = f(3x)$ and the positive axes.

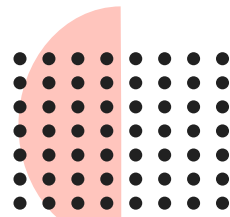
Dilation by factor $\frac{1}{3} \Rightarrow$ Area must be a third the original

6



04

Question Approach



CAS Setup

My ideal CAS setup:

Nothing... just a fresh document

All of the algorithms like Newton's Method, Bisection Method, etc are handled by UDFs, so I never had any notes pages.

Just make sure your CAS isn't in sleep mode and is readily open and won't have to boot from shut down before any assessment.

Calculation Mode:

Auto

CAS Mode:

Auto

Exact

Approximate

Usually Auto mode will give you the results you need, but sometimes calculations run in Auto will yield approximate solutions when you require an exact one.

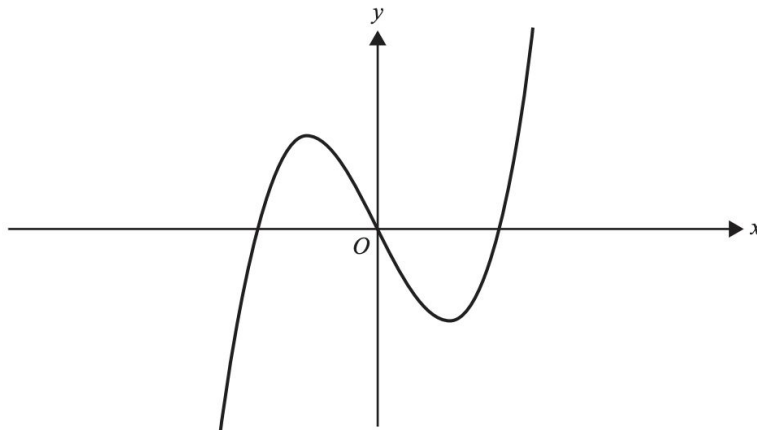
You can either switch to Exact when necessary, or always work in Exact.

Be careful however: Exact has vastly more potential to crash the CAS. Always use 'ctrl + enter' when an approximate solution is required.

Question 1 (10 marks)

Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 - px$, where $p \in \mathbb{R}$.

Part of the graph of f is shown below, when $p = 3$.



- a. Find the values of the x -intercepts of f , when $p = 3$.

1 mark

$$x = -\sqrt{3}, 0, \sqrt{3}$$

- b. Use the derivative f' to find the coordinates of the turning points of f , when $p = 3$.

2 marks

$$f'(x) = 3x^2 - 3$$

$$\rightarrow f'(x) = 0$$

$$\Rightarrow \text{Stps at: } (-1, 2), (1, -2)$$

c. i. Find the value of p for which f would have exactly one stationary point.

1 mark

$$p = 0$$

ii. Find the values of p for which f would not have any stationary points.

1 mark

$$p < 0$$

Thanks!

Feel free to reach out!
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